

RepoChat CICD

Pipeline Overview:

The pipeline has two independent tracks, backend and frontend, that only run when their respective directories change. Both tracks run in parallel when triggered.

On Every Push (any branch):

Detect Changes → backend/ changed? → backend/ unchanged? (skip)

If backend/ changed:

- Lint - ESLint checks for code issues
- Type check - TypeScript verifies all types
- Build - compiles TypeScript to JavaScript
- Unit tests - business logic tested in isolation
- Integration tests - real in-memory MongoDB, OpenAI mocked

If frontend/ changed:

- Lint - ESLint with React and Next.js rules
- Type check - TypeScript verifies all component types
- Build - next build confirms the static export compiles cleanly

On Every Merge or Push to Main (if CI passes):

If backend/ changed:

- Build TypeScript
- Authenticate to AWS via GitHub OIDC
- Build Docker image → push to Amazon ECR → update Lambda → wait for rollout

If frontend/ changed:

- Run next build → sync output to S3 (old files deleted) → invalidate CloudFront cache

After either deploy completes:

- E2E tests run against the live production URL using Playwright

Security:

- No AWS credentials are stored in GitHub
- Authentication uses GitHub OIDC - GitHub proves its identity to AWS and receives temporary credentials that expire after one hour
- All secrets are stored in AWS SSM Parameter Store and read directly by Lambda at startup - they never pass through the pipeline